CSE 111 Fall 2026: Project Checkpoint 1

Manufactorio

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Synopsis

Manufactorio is a database-ed manufacturing, planning, and simulation game using SQL as the framework to track all of the data, transformations, meanings, and calculation. The game has the player take control of an industrial manufacturing company on a foreign planet where they exploit resources in different biomes in order to set up production lines to create increasingly more complex products out of said resources. The database will track and control data such as which resources are being extracted and where, how much of a product the player is making, and what they are selling said product for. After listening for some simple user inputs before each iteration or "cycle," the database will update accordingly to allocate resources, funding, or product to be sold. By the end of a game the player can run some prefabricated queries to see complex analysis of their work such as a full line of production from resource to product or income estimates.

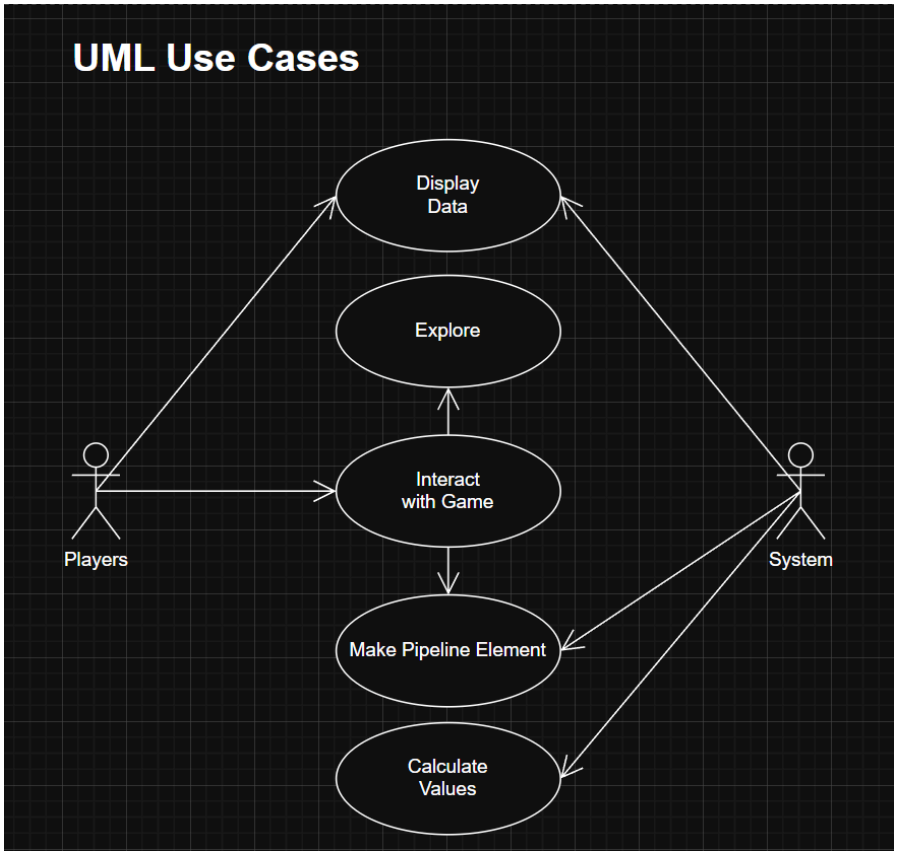
Authors & Contribution: Ri'chard 100%

UML Use Case Diagram

The main use case of the database will be directly providing the game's data to the player and the game by managing the underlying content of the fictional world. The player will use the database to make informed decisions in the game and the game will use the database as a lookup for relevant information.

Authors & Contribution: Ri'chard 100%

Use Case Diagram:

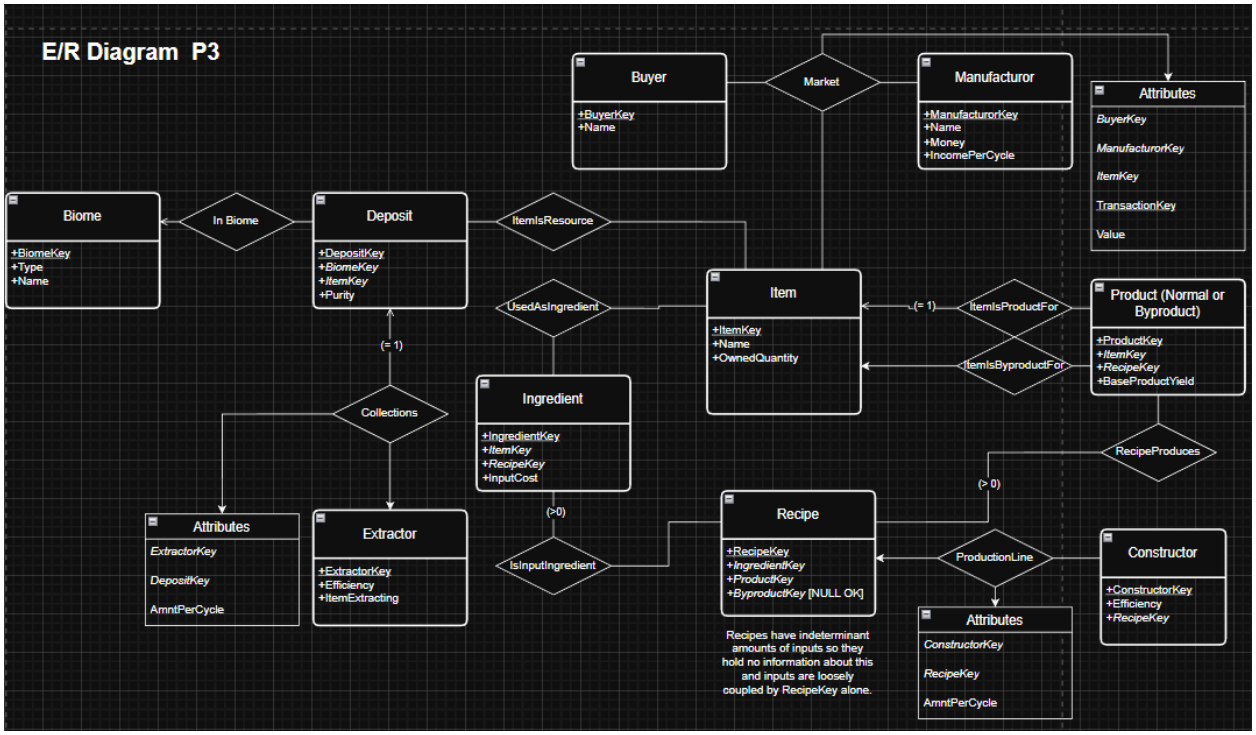


Authors & Contribution: Ri'chard 100%

ER Diagram

The ER Diagram is provided below.

ERD-1:



Authors & Contribution: Ri'chard 100%

Relational Schema

Relational schemas along with their descriptions.

[Relation 1] - Biome

Schema: Biome(biome_key, type, size, name)

Primary Key: biome_key

Foreign Keys: None

Description:

- Represents each biome, including their type, size, and name.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 2] - Deposit

Schema: Deposit(deposit_key, biome_key, item_key, purity)

Primary Key: deposit_key

Foreign Keys: biome_key (Biome), item_key (Item)

Description:

- Represents resource deposits available in specific biomes, including the item the deposit contains and its purity.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 3] - Extractor

Schema: Extractor(extractor_key, item_extracting, efficiency)

Primary Key: extractor_key

Foreign Keys: None

Description:

- Represents available extractors and what item they extract, along with extraction efficiency.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 4] - Constructor

Schema: Constructor(constructor_key, efficiency)

Primary Key: constructor_key

Foreign Keys: None

Description:

- Represents available constructors and their efficiency.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 5] - Item

Schema: Item(item_key, name, quantity)

Primary Key: item_key

Foreign Keys: None

Description:

- Represents available items, including their names and quantities.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 6] - Ingredient

Schema: Ingredient(ingredient_key, item_key, recipe_key, cost)

Primary Key: ingredient_key

Foreign Keys: item_key (Item), recipe_key (Recipe)

Description:

- Represents an ingredient used in a specific recipe and its cost.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 7] - Product

Schema: Product(product_key, item_key, recipe_key, base_yield)

Primary Key: product_key

Foreign Keys: item_key (Item), recipe_key (Recipe)

Description:

- Represents the product of a specific recipe, including its base yield.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 8] - Recipe

Schema: Recipe(recipe_key, ingredient_key, product_key, byproduct_key)

Primary Key: recipe_key

Foreign Keys: ingredient_key (Ingredient), product_key (Product), byproduct_key (Product)

Description:

- Represents available recipes, including their required ingredient, resulting product, and optional byproduct.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 9] - Buyer

Schema: Buyer(buyer_key, name)

Primary Key: buyer_key

Foreign Keys: None

Description:

- Represents different buyers, including their name.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 10] - Manufacturer

Schema: Manufacturer(manufacturer_key, name, money, income_per_cycle)

Primary Key: manufacturer_key

Foreign Keys: None

Description:

- Represents different manufacturers, including their name, money, and income per cycle.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 11] - Extractions

Schema: Extractions(extraction_key, extractor_key, deposit_key, amnt_per_cycle)

Primary Key: extraction_key

Foreign Keys: extractor_key (Extractor), deposit_key (Deposit)

Description:

- Represents extractions, including the extractor involved, the deposit the extraction is taking place, and the amount of resources gained per cycle.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 12] - Production

Schema: Production(production_key, constructor_key, recipe_id, amnt_per_cycle)

Primary Key: production_key

Foreign Keys: constructor_key (Constructor), recipe_id (Recipe)

Description:

- Represents production, including the constructor and recipe being used and the amount created per cycle.

Authors & Contribution: Ri'chard 20%, Joshua 80%

[Relation 13] - Market

Schema: Market(transaction_key, manufacturer_key, buyer_key, item_key, value)

Primary Key: transaction_key

Foreign Keys: manufacturer_key (Manufacturer), buyer_key (Buyer), item_key (Item)

Description:

- Represents transactions at the market, including the buyer and manufacturer involved, the item being bought, and the transaction value in dollars.

Authors & Contribution: Ri'chard 20%, Joshua 80%

Core Entities	Key Relationships and Cardinalities	Resulting Relations (Entity + Junction Tables)
Biome Deposit Extractor Constructor Buyer Manufacturer Item Recipe Product Ingredient	Biomes - Depost [M:N] (InBiome) **Deposit - Extractor [1:1] (Collections)** [1:1] (ItemIsResource) Item - Ingredient [1:1] (UsedAsIngredient) Ingredient - Recipe [M:N] (IsInputIngredient) **Buyer - Manufacturer - Item [M:N:O] (Market)** Item - Product [1:M] (ItemIsProductFor) Item - Product [1:M] (ItemIsByproductFor) Recipe - Product [1:1] (RecipeProduces) **Recipe - Constructor [1 : M] (ProductionLine)**	**10 Entity Tables:** Biome Deposit Item Recipe Ingredient Product Extractor Constructor Buyer Manufacturer **3 Relation Tables:** Market (Manufacturers:Items:Buyers) Collections (Deposits:Biomes) ProductionLine (Recipe:Constructor)

Relations in Bold have tables with unique attributes and others do not.